Vol. **000**, 1–9 (2026) Preprint 9 November 2026 Compiled using LaTeX style file v3.0

Halo mass functions from maximum entropy distributions in collisionless dark matter flow

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9 November 2026

ABSTRACT

In a self-gravitating collisionless system the velocity distribution of all particles that maximises the system entropy, the X distribution, is a scale mixture of the Gaussian velocity distributions of halo groups. We use this property to connect the halo mass function with the maximum entropy distributions. The distribution H of the virial dispersion carried by the particles is equivalent to the halo mass function, and it is related to X, to the distribution P of the squared velocity and to the distribution J of the number of halos by integral transforms. Without halo motion, the mixing distribution is exactly the generalized inverse Gaussian distribution H_∞. It is a maximum entropy distribution for a potential exponent $n = 0$, but not for gravity. For small halos, H reduces to the Press–Schechter form. We test these relations with the friends-of-friends halos of a cosmological N-body simulation at $z = 0$. The mixture of Gaussians built from the measured dispersions of the halo particles reproduces their velocity distribution to 0.03 dex (rms over $|v| < 6\sigma_0$) and its kurtosis (5.10 against 5.12). However, no halo particle has a total dispersion below $0.51\,\sigma_0^2$, whereas the maximum entropy mixture places 0.28–0.34 of the particles there. We prove that, for any halo velocity dispersion bounded away from zero, the exact equation for the mass function has no solution. Its push-forward solution leaves a fraction 0.27 of the mixture unassigned. The simulated mass function in $\nu = \sigma_v^2/\bar{\sigma}_h^2$ is described by the two-parameter P form with rms 0.18 in $\ln f$, comparable to fitted Sheth–Tormen and double-λ forms. It is not independent of redshift: the mean ν rises from 0.56 at $z = 2$ to 1.14 at $z = 0$. All data, code and results are published with the paper.

Key words: dark matter – large-scale structure of Universe – methods: numerical – methods: statistical

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1 INTRODUCTION

The halo mass function, the distribution of mass among halos of different sizes, is a basic prediction of structure formation and one of the main links between the dark matter model and observations. The abundance of galaxy clusters, measured in X-ray, optical and Sunyaev–Zel'dovich surveys, constrains the amplitude of matter fluctuations, the matter density and the equation of state of dark energy through the mass function [1, 2, 3, 4]. The same function underlies the halo model of galaxy clustering [5, 6] and the abundance matching that connects galaxies to their halos [7, 8, 9]. Cluster cosmology needs the mass function to within a few per cent, and whether it is universal, a single function of the peak height at all redshifts and in all cosmologies, depends on the halo definition and is still discussed [10, 11, 12].

Most theories of the mass function start from the linear density field. The Press–Schechter (PS) formalism [13] and its excursion-set derivation [14, 15] predict the mass function from the variance of the linear density field and a collapse threshold δ_c given by spherical collapse [16] or, equivalently, by the two-body collapse model [17]. In terms of $\nu = \delta_c^2/\sigma_\delta^2(m_h)$, with $\sigma_\delta^2(m_h)$ the variance of the linear density field smoothed on the scale of mass m_h, the PS mass function reads

$$f_{PS}(\nu) = \frac{1}{\sqrt{2\pi\nu}}\, e^{-\nu/2}. \tag{1}$$

Ellipsoidal collapse with a moving barrier gives the Sheth–Tormen

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(ST) form [18, 19, 20],

$$f_{ST}(\nu) = A\sqrt{\frac{2q}{\pi}}\left[1+(q\nu)^{-p}\right]\frac{e^{-q\nu/2}}{2\sqrt{\nu}},$$
$$A = \frac{\sqrt{\pi}}{\Gamma(1/2)+2^{-p}\Gamma(1/2-p)}, \tag{2}$$

with $A = 0.3222$ for $p = 0.3$ and $q = 0.75$. Fits to simulations provide empirical forms [10, 21, 22, 23], for example

$$f_{JK}(\nu) = \frac{0.315}{2\nu}\exp\left[-\left|\ln\left(\sqrt{\nu}/\delta_c\right)+0.61\right|^{3.8}\right], \tag{3}$$

which is the fit of Jenkins *et al.* [21] written for ν ($d\nu = 2\nu\, d\ln\sigma_\delta^{-1}$); it is not normalised ($\int_0^\infty f_{JK}d\nu = 0.57$). The inverse mass cascade gives the double-λ mass function [24, 25],

$$f_{D\lambda}(\nu) = \frac{(4\eta_0)^{-q/2}}{\Gamma(q/2)}\,\nu^{q/2-1}\exp\left(-\frac{\nu}{4\eta_0}\right), \tag{4}$$

with $\eta_0 = 0.76$ and $q = 0.556$ for the simulation used here. In that picture the mass function results from a random walk of halos in mass space; a unified random-walk theory of mass functions and density profiles is given in Xu [26].

A second route starts from statistical mechanics. A self-gravitating collisionless system with long-range interactions maximises its entropy by forming halos of many sizes, and the velocity distribution of all particles that maximises the entropy is the X distribution [27]; see Lynden-Bell [28], White and Narayan [29], Padmanabhan [30], Hjorth and Williams [31] for earlier work and Jaynes [32] for the principle. Because the particles of one halo group are close to Gaussian, the X distribution is a scale mixture of Gaussians, and the mixing distribution is fixed by the halo population. The two routes have not been connected: it is not known what the maximum entropy distributions imply for the halo mass function, whether that implication is borne out by simulated halos, and whether it can explain why the mass function is nearly universal.

In this paper we connect them and test every step with the friends-of-friends halos of a cosmological N-body simulation. We show that the distribution H of the virial dispersion carried by the particles is equivalent to the halo mass function and is related to X by an integral transform. Without halo motion the mixing distribution is exactly the generalized inverse Gaussian distribution H_∞; it is a maximum entropy distribution for a potential exponent $n = 0$, but not for gravity, and for small halos H reduces to the PS form. In the simulation, the mixture of Gaussians built from the measured dispersions of the halo particles reproduces their velocity distribution to 0.03 dex, but no halo particle has a total dispersion below $0.51\,\sigma_0^2$, where the maximum entropy mixture places 0.28–0.34 of the particles. We prove that the exact equation for the mass function has no solution when halos move with a velocity dispersion bounded away from zero; its push-forward solution leaves a fraction 0.27 of the mixture unassigned. The simulated mass function is described by a two-parameter form with rms 0.18 in $\ln f$, comparable to fitted ST and double-λ forms, and it is not independent of redshift: the mean ν rises from 0.56 at $z = 2$ to 1.14 at $z = 0$.

The paper is organized as follows. Section 2 defines the distributions and their relations, and Section 3 treats the limits of large and small halos. Section 4 compares the predictions with the simulation, Section 5 discusses the complete equation for the mass function, and Section 6 concludes.

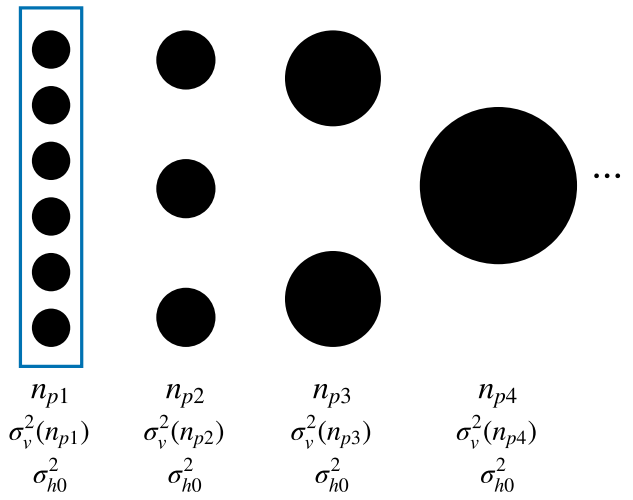


Figure 1. Halo groups. Halos are sorted by the number of particles n_p; every group has a virial dispersion $\sigma_v^2(n_p)$ and a halo velocity dispersion σ_h^2 that depends only weakly on n_p (Fig. 2).

2 HALO GROUPS AND LIMITING DISTRIBUTIONS

2.1 Halo groups

Consider N collisionless particles interacting through a two-body potential $V(r) \propto r^n$ ($n = -1$ for gravity). In the halo description every particle belongs to one halo; halos of the same size n_p (number of particles) form a halo group (Fig. 1). A group is characterised by the halo virial dispersion $\sigma_v^2(n_p)$, the mean over its halos of the 1D variance of the particle velocities about the halo mean velocity, and by the halo velocity dispersion $\sigma_h^2(n_p)$, the 1D variance of the halo mean velocities [27]. Virial equilibrium with $m_h \propto r_h^3$ gives

$$\sigma_v^2 \propto \frac{Gm_h}{r_h^{-n}} \propto m_h^{1+n/3} = m_h^{1/\beta}, \qquad \beta = \frac{3}{3+n}, \tag{5}$$

and the 1D velocity dispersion of all particles of a group is

$$\sigma^2(n_p) = \sigma_v^2(n_p) + \sigma_h^2(n_p). \tag{6}$$

The whole system has the 1D velocity dispersion σ_0^2.

2.2 Distributions and their relations

We use six distributions: $X(v)$ of the 1D particle velocity, $Z(v)$ of the speed, $E(\varepsilon)$ of the particle energy [27], and, introduced here, $H(\sigma_v^2)$, the fraction of particles per unit virial dispersion (each particle carries the virial dispersion of its halo), $J(\sigma_v^2)$, the fraction of halos per unit virial dispersion, and $P(x)$, the distribution of $x = v^2$. If the particle velocities of a group are Gaussian with variance σ^2 (Eq. 6),

$$X(v) = \int_0^\infty \frac{e^{-v^2/2\sigma^2}}{\sqrt{2\pi}\,\sigma} H(\sigma_v^2)\,d\sigma_v^2, \tag{7}$$

so that X is a scale mixture of Gaussians whose mixing variable is the total dispersion σ^2; for $\sigma_h^2 = 0$ the mixing variable is σ_v^2 itself. Moments and transforms follow directly:

$$\int_{-\infty}^\infty X v^2 dv = \int_0^\infty H\sigma^2 d\sigma_v^2 = \langle\sigma_v^2\rangle + \langle\sigma_h^2\rangle = \sigma_0^2,$$
$$\bar\sigma_h^2 \equiv \langle\sigma_h^2\rangle = \int_0^\infty H\sigma_h^2 d\sigma_v^2, \tag{8}$$

$$\int_{-\infty}^{\infty} X(v)e^{-vt}dv = \int_0^{\infty} H(\sigma_v^2)\,e^{\sigma^2 t^2/2}d\sigma_v^2,$$
$$\int_{-\infty}^{\infty} Xv^m dv = \frac{2^{m/2}}{\sqrt{\pi}}\Gamma\!\left(\frac{m+1}{2}\right)\int_0^{\infty} H\sigma^m d\sigma_v^2 \quad (m \text{ even}). \tag{9}$$

Counting the particles of the groups with σ_v^2 in $[\sigma_v^2, \sigma_v^2 + d\sigma_v^2]$ in two ways relates H and J,

$$H(\sigma_v^2) = J(\sigma_v^2)\frac{n_p(\sigma_v^2)}{\bar N}, \qquad \bar N = \int_0^{\infty} J\,n_p\,d\sigma_v^2, \tag{10}$$

with $\bar N$ the mean number of particles per halo. The distribution of $x = v^2$ is

$$P(x) = \frac{X(\sqrt{x})}{\sqrt{x}} = \int_0^{\infty}\frac{e^{-x/2\sigma^2}}{\sqrt{2\pi x}\,\sigma}H(\sigma_v^2)\,d\sigma_v^2,$$
$$\int_0^{\infty} Pe^{-xt}dx = \int_0^{\infty}\frac{H\,d\sigma_v^2}{\sqrt{1+2\sigma^2 t}}, \tag{11}$$
$$\int_0^{\infty} Px^m dx = \frac{2^m}{\sqrt{\pi}}\Gamma\!\left(m+\tfrac{1}{2}\right)\int_0^{\infty} H\sigma^{2m}d\sigma_v^2,$$

where the kernel is the χ^2 distribution of one degree of freedom scaled by σ^2.

The maximum entropy velocity distribution is [27]

$$X(v) = \frac{e^{-\sqrt{\alpha^2+(v/v_0)^2}}}{2\alpha v_0 K_1(\alpha)}, \qquad \sigma_0^2 = \alpha\frac{K_2(\alpha)}{K_1(\alpha)}v_0^2, \tag{12}$$

with K_ν the modified Bessel function of the second kind, a shape parameter α and a velocity scale v_0. It has a Gaussian core of variance αv_0^2 for $|v| \ll v_0$ and exponential wings $\propto e^{-|v|/v_0}$ for $|v| \gg v_0$; it is Laplace for $\alpha \to 0$ and Gaussian for $\alpha \to \infty$. Its moment-generating function is $K_1(\alpha\sqrt{1-v_0^2t^2})/[K_1(\alpha)\sqrt{1-v_0^2t^2}]$ for $|t| < 1/v_0$ and its even moments are $M_X(m) = (2\alpha)^{m/2}\Gamma[(1+m)/2]K_{1+m/2}(\alpha)v_0^m/[\sqrt{\pi}K_1(\alpha)]$. Equation (12) is the symmetric hyperbolic distribution, which is a normal variance mixture [33]. Combining Eqs. (9) and (12),

$$\int_0^{\infty} H(\sigma_v^2)\,e^{-\sigma^2 t}d\sigma_v^2 = \frac{K_1\!\left(\alpha\sqrt{1+2v_0^2t}\right)}{K_1(\alpha)\sqrt{1+2v_0^2t}}, \tag{13}$$

whose inverse, as a distribution of the total dispersion σ^2, is the generalized inverse Gaussian (GIG) distribution with index 1 [34],

$$H_\infty(\sigma^2) = \frac{1}{2\alpha v_0^2 K_1(\alpha)}\exp\left[-\frac{\alpha}{2}\left(\frac{\sigma^2}{\alpha v_0^2}+\frac{\alpha v_0^2}{\sigma^2}\right)\right]. \tag{14}$$

The distribution of $x = v^2$ is $P(x) = e^{-\sqrt{\alpha^2+x/v_0^2}}/[2\alpha v_0 K_1(\alpha)\sqrt{x}]$, with moments $M_P(m) = K_{1+m}(\alpha)\Gamma(m+1/2)(2\alpha v_0^2)^m/[\sqrt{\pi}K_1(\alpha)]$. Table 1 lists the family for $n = 0, -1, -2$; for $n = 0$ ($\alpha \to 0$, no halo motion) $H(\sigma_v^2) = e^{-\sigma_v^2/\sigma_0^2}/\sigma_0^2$, and for $n = -2$ ($\alpha \to \infty$) all particles have $\sigma_v^2 = 0$. The statistical properties of H_∞ and P are collected in Table A1; every entry has been checked numerically.

3 LIMITS OF LARGE AND SMALL HALOS

3.1 Large halos: H_∞ and J_∞

For large halos $\sigma_v^2 \gg \sigma_h^2$, and with $\sigma_h^2 = 0$ the mixing distribution of the virial dispersion is $H_\infty(\sigma_v^2)$ of Eq. (14). Its moment-generating function, moments and entropy are

$$M_{H_\infty}(t) = \frac{K_1\!\left(\alpha\sqrt{1-2v_0^2t}\right)}{K_1(\alpha)\sqrt{1-2v_0^2t}}, \qquad \langle(\sigma_v^2)^m\rangle = \frac{K_{1+m}(\alpha)}{K_1(\alpha)}(\alpha v_0^2)^m,$$
$$S_{H_\infty} = -\int_0^{\infty} H_\infty \ln H_\infty\,d\sigma_v^2 = \alpha\frac{K_2(\alpha)}{K_1(\alpha)} - 1 + \ln\left[2\alpha v_0^2 K_1(\alpha)\right]. \tag{15}$$

With the virial mass–dispersion relation $n_p = \mu\bar N(\sigma_v^2/v_0^2)^\beta$ (Eq. 5) and Eq. (10), the number distribution of halos is

$$J_\infty(\sigma_v^2) = \frac{H_\infty}{\mu(\sigma_v^2/v_0^2)^\beta}$$
$$= \frac{1}{2\alpha v_0^2 K_{\beta-1}(\alpha)}\left(\frac{\alpha v_0^2}{\sigma_v^2}\right)^\beta \exp\left[-\frac{\alpha}{2}\left(\frac{\sigma_v^2}{\alpha v_0^2}+\frac{\alpha v_0^2}{\sigma_v^2}\right)\right],$$
$$\mu = \alpha^{-\beta}\frac{K_{\beta-1}(\alpha)}{K_1(\alpha)}, \tag{16}$$

a GIG distribution with index $1-\beta$, with moments $K_{m+1-\beta}(\alpha)(\alpha v_0^2)^m/K_{1-\beta}(\alpha)$ and moment-generating function $K_{\beta-1}(\alpha\sqrt{1-2v_0^2t})/[K_{\beta-1}(\alpha)(1-2v_0^2t)^{(1-\beta)/2}]$; the mass–dispersion relation becomes $n_p = \bar N K_{\beta-1}(\alpha)/K_1(\alpha)\,(\sigma_v^2/\alpha v_0^2)^\beta$. For $\sigma_v^2 \to \infty$, H_∞ decays as $e^{-\sigma_v^2/2v_0^2}$.

3.2 Maximum entropy characterisation

Maximising $S[H] = -\int H\ln H$ under the three constraints $\int H = 1$, $\int H\sigma_v^2 = \langle\sigma_v^2\rangle$ and $\int J = \int H/[\mu(\sigma_v^2/v_0^2)^\beta] = 1$ (the normalisation of J) gives

$$H(\sigma_v^2) = e^{\lambda_1 - 1}\exp\left[\lambda_2\sigma_v^2 + \frac{\lambda_3}{\mu}\left(\frac{v_0^2}{\sigma_v^2}\right)^\beta\right]. \tag{17}$$

For $\beta = 1$ ($n = 0$), the multipliers $\lambda_1 = 1 - \ln[2\alpha v_0^2 K_1(\alpha)]$, $\lambda_2 = -1/2v_0^2$ and $\lambda_3 = -\mu\alpha^2/2$ give H_∞, and for $\alpha \to 0$ the exponential distribution of Table 1. For gravity ($\beta = 3/2$) the stationary point is $\exp[\lambda_2\sigma_v^2 + c(\sigma_v^2)^{-3/2}]$, which is not H_∞. The maximum entropy property of the large-halo limit therefore holds for $n = 0$ only; for $n = -1$, H_∞ is the mixing distribution implied by the maximum entropy X distribution, but not itself a maximum entropy distribution under these constraints.

3.3 Small halos: H_s and the Press–Schechter form

Particles with small v^2 reside mostly in small halos with small σ_v^2. Approximating the χ^2 kernel of Eq. (11) by $\delta(x - \sigma_v^2)$ for small σ^2 gives $H(x) \approx P(x)$, and using the Gaussian core of X,

$$H_s(\sigma_v^2) = \frac{1}{\sqrt{2\pi\alpha v_0^2\sigma_v^2}}\exp\left(-\frac{\sigma_v^2}{2\alpha v_0^2}\right), \tag{18}$$

a χ^2 distribution of one degree of freedom with mean αv_0^2. The approximation neglects σ_h^2, which does not vanish for small halos (Fig. 2). With

$$\nu = \frac{\sigma_v^2(m_h)}{\bar\sigma_h^2} = \left(\frac{m_h}{m_h^*}\right)^{2/3}, \qquad f(\nu) = \bar\sigma_h^2 H(\nu\,\bar\sigma_h^2), \tag{19}$$

Table 1. Parameters and distributions for typical potential exponents n. The values for $n=-1$ in brackets are those of the simulation (σ_0^2 of the halo particles).

n	β	α	v_0^2	$\langle\sigma_h^2\rangle$	$\langle\sigma_v^2\rangle$	$X(v)$	$H(x=\sigma_v^2)$	$P(x=v^2)$
0	1	0	$\sigma_0^2/2$	0	σ_0^2	$e^{-\sqrt{2}\lvert v\rvert/\sigma_0}/(\sqrt{2}\sigma_0)$	$e^{-x/\sigma_0^2}/\sigma_0^2$	$e^{-\sqrt{2x}/\sigma_0}/(\sigma_0\sqrt{2x})$
-1	3/2	free	$\sigma_0^2 K_1/(\alpha K_2)$	$\approx\sigma_0^2 K_1/K_2$ [$0.459\sigma_0^2$]	$\approx(1-K_1/K_2)\sigma_0^2$ [$0.525\sigma_0^2$]	Eq. (12)	Eq. (14) for $\sigma_h=0$	$P(x)$ above
-2	3	∞	0	σ_0^2	0	$e^{-v^2/2\sigma_0^2}/(\sqrt{2\pi}\sigma_0)$	$\delta(x)$	$e^{-x/2\sigma_0^2}/(\sigma_0\sqrt{2\pi x})$

Table 2. Parameters of the N-body simulation.

Run	Ω_0	Λ	h	σ_8	L (Mpc/h)	N	m_p ($M_\odot/h$)	$l_{\rm soft}$ (kpc/h)
SCDM1	1.0	0.0	0.5	0.51	239.5	256^3	2.27×10^{11}	36

where m_h^* is defined by $\sigma_v^2(m_h^*)=\bar\sigma_h^2$, Eq. (18) becomes

$$f_{H_s}(\nu)=\frac{1}{\sqrt{2\pi\gamma\nu}}\exp\left(-\frac{\nu}{2\gamma}\right),\qquad \gamma=\frac{\alpha v_0^2}{\bar\sigma_h^2}, \tag{20}$$

the PS function for $\gamma=1$. Since $\alpha v_0^2\approx\langle\sigma_h^2\rangle$ [27, Eq. 37], $\gamma\approx1$; in the simulation $\gamma=0.97$ ($\alpha=1.33$ and σ_0^2 of the halo particles). The identification of ν with $\delta_c^2/\sigma_\delta^2$ is a proportionality that holds when $\sigma_\delta^2\propto m_h^{-2/3}$, i.e. for an effective spectral index -1 on halo scales. In the same variable, P and H_∞ read

$$\begin{aligned} P(\nu)&=\frac{e^{-\sqrt{\alpha^2+\alpha\nu/\gamma}}}{2K_1(\alpha)\sqrt{\alpha\gamma\nu}},\\ f_{H_\infty}(\nu)&=\frac{1}{2\gamma K_1(\alpha)}\exp\left[-\frac{\alpha}{2}\left(\frac{\nu}{\gamma}+\frac{\gamma}{\nu}\right)\right]. \end{aligned} \tag{21}$$

4 COMPARISON WITH AN N-BODY SIMULATION

4.1 Data

We use the Virgo consortium SCDM simulation [35, 36] (Table 2); the Virgo data have also been used to test models of halo abundances [19]. Halos are identified by friends-of-friends with a comoving linking length of $0.2\,{\rm Mpc}/h$ ($b=0.214$ of the mean interparticle separation). At $z=0$, 10,290,483 particles (61.3 per cent) are in 1,674,583 halos with $n_p\geqslant2$ and 6,486,733 are in no halo; distributions of halo quantities below are normalised to the halo particles. Group statistics at ten redshifts are those of the public dataset [37]; per-halo and per-particle quantities are computed from the halo catalogue. Velocities are in units of $u_0=354.6\,{\rm km\,s^{-1}}$, the 1D rms velocity of all particles at $z=0$; the halo particles have $\sigma_0^2=1.242u_0^2$.

4.2 Virial and halo velocity dispersions

Figure 2 shows σ_v^2, σ_h^2 and σ^2 against halo size at $z=0$. Fits that describe the data are

$$\sigma_v^2=0.03\,n_p^{2/3}u_0^2,\qquad \sigma_h^2=0.375\left[1-\tanh\left(\frac{n_p-500}{600}\right)\right]u_0^2, \tag{22}$$

with $\sigma_v^2\propto m_h^{2/3}$ as expected for $n=-1$ (Eq. 5). The halo velocity dispersion decreases slowly up to $n_p\approx500$ and then falls steeply. Its particle-weighted mean is $\bar\sigma_h^2=0.570u_0^2$ and the smallest halos have $\sigma_h^2=0.642$ ($n_p=2$) and $0.655u_0^2$ ($n_p=3$), so that σ_h^2 is bounded away from zero for all but the largest halos. The scale m_h^* ($\sigma_v^2=\bar\sigma_h^2$) is at $n_p^*\approx86$, i.e. $m_h^*\approx2.0\times10^{13}M_\odot/h$.

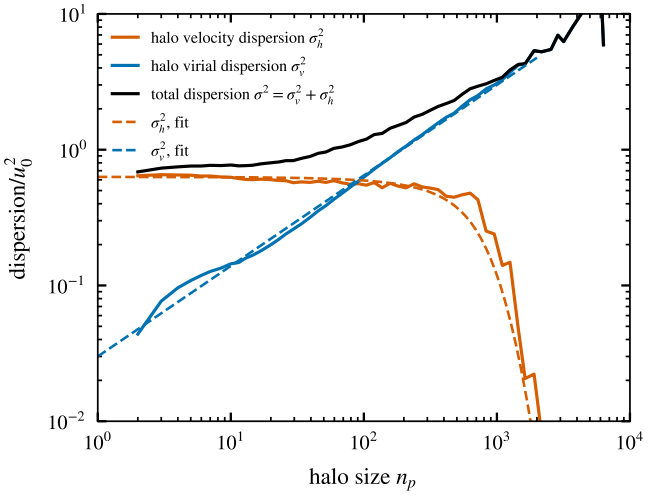


Figure 2. Halo velocity dispersion σ_h^2, virial dispersion σ_v^2 and total dispersion $\sigma^2=\sigma_v^2+\sigma_h^2$ against halo size at $z=0$ (particle-weighted averages over the groups in bins of n_p). Dashed: fits of Eq. (22).

4.3 A direct test of the mixture relation

Each halo particle carries a total dispersion $\sigma^2=\sigma_{vh}^2+\sigma_h^2(n_p)$, the virial dispersion of its halo plus the halo velocity dispersion of its group (both 1D, about zero). Their mean is exactly σ_0^2 (Eq. 8). Figure 3a shows their distribution together with the mixing distribution $H_\infty(\sigma^2)$ of the X distribution with variance σ_0^2. For $\sigma_0^2<\sigma^2<4\sigma_0^2$ the two agree to within 0.35 dex (beyond, few halos contribute and the histogram is noisy), but the measured distribution has a sharp lower edge: no halo particle has $\sigma^2<0.51\sigma_0^2$, whereas H_∞ places 0.28 ($\alpha=1.33$) to 0.34 ($\alpha=0.72$) of the particles there. The halo population realises a different mixture from the maximum entropy one.

Nevertheless the mixture of Gaussians built with the measured σ^2, Eq. (7), reproduces the velocity distribution of the halo particles (Fig. 3b): the rms difference is 0.03 dex for $|v|<6\sigma_0$ and the kurtosis is 5.10 against 5.12 measured. The X distribution with $\alpha=1.33$, fitted to the velocities of particle pairs at $0.1\,{\rm Mpc}/h$ [27], underestimates the wings beyond $5\sigma_0$; $\alpha=0.72$, which matches the kurtosis of the halo particles, is closer. Two lessons follow. The Gaussian-mixture picture behind Eqs. (7)–(11) is well supported. But the velocity distribution constrains the mixing distribution only weakly: mixtures that differ substantially at small σ^2 give similar X, so the mass function cannot be recovered from X alone; it requires the halo velocity dispersion explicitly (Section 5).

4.4 The mass function

The simulated H is obtained by giving each particle the virial dispersion σ_{vh}^2 of its halo and histogramming over all halo particles; $f(\nu)$ follows from Eq. (19) with $\bar\sigma_h^2=0.570u_0^2$ (Fig. 4). The ST and

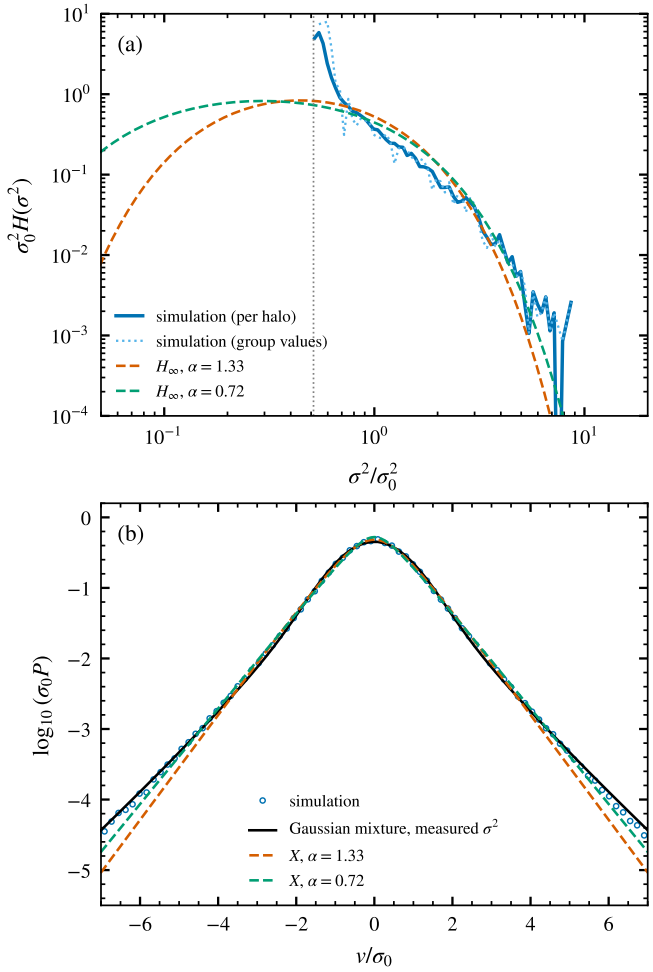


Figure 3. Test of the mixture relation at $z = 0$. (a) Distribution of the total dispersion σ^2 carried by the halo particles (per-halo virial dispersion plus group halo velocity dispersion; dotted: group virial dispersion) and the mixing distribution H_∞ of X with variance σ_0^2 for $\alpha = 1.33$ and 0.72. The vertical line marks the smallest σ^2 of any halo particle. (b) Velocity distribution of the halo particles, the mixture of Gaussians with the measured σ^2 (Eq. 7) and the X distribution.

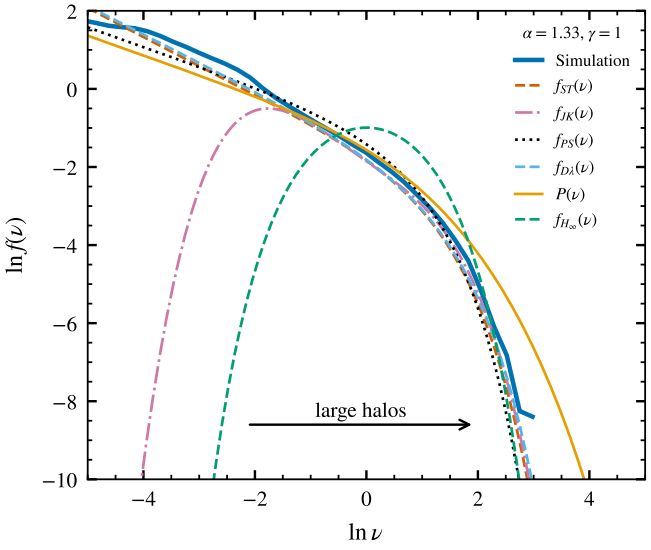


Figure 4. Halo mass function $f(\nu)$ at $z = 0$ from the H distribution of the simulation (thick), compared with PS, ST, Jenkins et al. (JK) and double-λ, and with $P(\nu)$ and $f_{H_\infty}(\nu)$ of Eq. (21) for $\alpha = 1.33$, $\gamma = 1$.

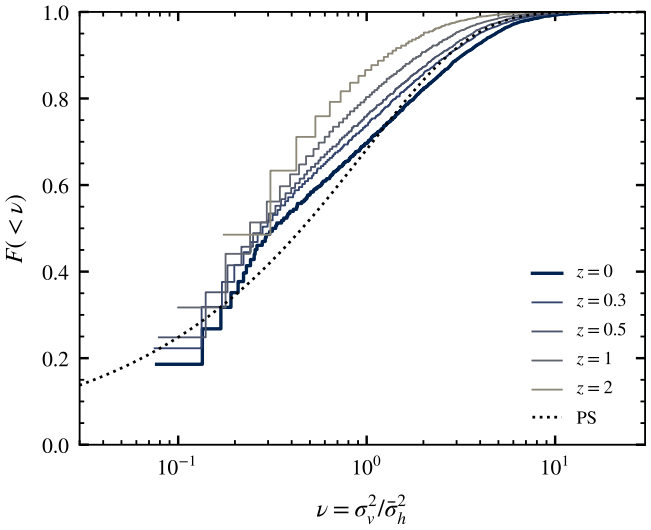


Figure 5. Cumulative fraction of the halo mass in groups with $\nu = \sigma_v^2/\bar{\sigma}_h^2(z)$ below the abscissa, for $z = 0$, 0.3, 0.5, 1 and 2 (group values), and the PS distribution.

double-λ forms with their standard parameters follow the simulation more closely than PS: the rms differences in $\ln f$ over $-4 < \ln\nu < 2.6$ are 0.27 (double-λ), 0.35 (ST) and 0.45 (PS). PS underestimates the mass in large halos (ratio of simulation to PS 2.00 at $\ln\nu = 2$), overestimates it for intermediate halos (0.80 at $\nu = 1$; $0.19 \lesssim \nu \lesssim 3.1$) and underestimates it again for the smallest halos (1.44 at $\ln\nu = -3$). The P distribution with $\alpha = 1.33$, $\gamma = 1$ follows the simulation to within 0.21 in $\ln f$ for $-1 < \ln\nu < 1$, lies 0.6 below it at $\ln\nu = -3$ and above it for $\ln\nu \gtrsim 1.5$. The large-halo limit f_{H_∞}, which neglects the halo velocity dispersion, contains no small halos and approaches the simulation only near $\ln\nu \approx 2$ (difference 0.2 in $\ln f$).

Fitting two-parameter forms to the simulation (Fig. 6) gives rms differences of 0.18 for $P(\nu)$ ($\alpha = 0.43$, $\gamma = 0.21$), 0.18 for double-λ ($\eta_0 = 0.79$, $q = 0.67$) and 0.21 for ST ($q = 0.68$, $p = 0.21$). The P form, derived here as the small-halo limit, is thus an adequate empirical description over the whole range, although its fitted parameters are far from the values expected from the velocity distribution, so the fit should not be read as a determination of α and γ.

4.5 Redshift dependence

If $f(\nu)$ were a stationary maximum entropy distribution with redshift-independent parameters, it would not depend on redshift. Figure 5 shows the cumulative fraction of the halo mass against $\nu = \sigma_v^2/\bar{\sigma}_h^2(z)$ for $z = 0$–2, from the group values of the public dataset. The distribution shifts steadily to larger ν as the universe evolves: the mean ν is 0.56, 0.72, 0.85, 0.94 and 1.14 at $z = 2$, 1, 0.5, 0.3 and 0, while the fraction of the mass in halos grows from 16 to 61 per cent. In this simulation the mass function in ν has therefore not reached a redshift-independent form by $z = 0$; whether it does so at later times, or in another scaling variable, remains open.

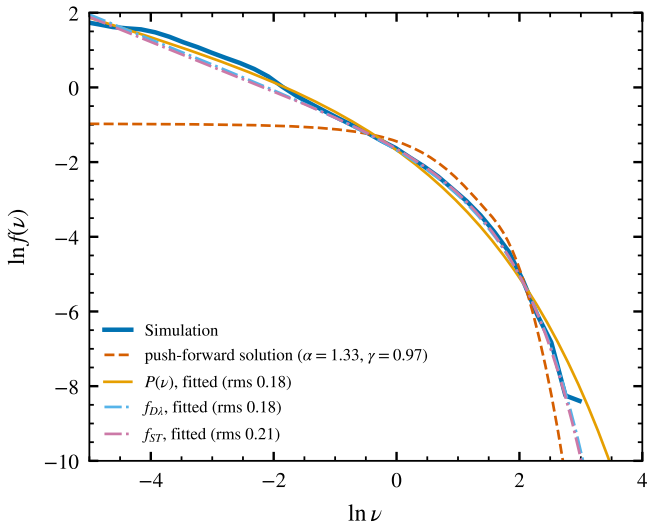


Figure 6. Simulated mass function (as Fig. 4) compared with the push-forward solution of Eq. (23) (Eq. 24; normalised to $1-D$) and with the fitted P, double-λ and ST forms (rms of $\ln f$ in brackets).

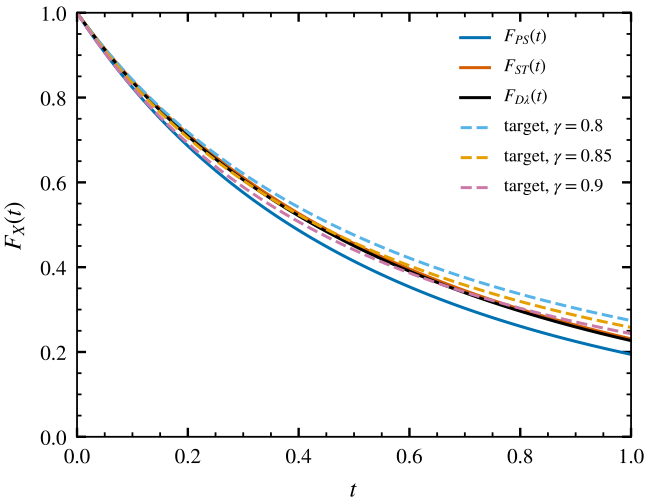


Figure 7. Transformed functions $F_X(t)$ of the PS, ST and double-λ mass functions (t is a transform variable, not time) with ν_h from Eq. (22) and $\bar\sigma_h^2 = 0.57u_0^2$, and the right-hand side of Eq. (23) for $\alpha = 4/3$ and $\gamma = 0.8$, 0.85, 0.9.

5 THE COMPLETE EQUATION FOR THE MASS FUNCTION

5.1 Equation and its exact properties

With $\nu_h = \sigma_h^2/\bar\sigma_h^2$, so that $\sigma^2/\bar\sigma_h^2 = \nu + \nu_h(\nu)$, Eqs. (9), (13) and (19) give

$$\begin{aligned}\int_0^\infty f(\nu)\, e^{-[\nu+\nu_h(\nu)]t}\, d\nu &= \frac{K_1\left(\alpha\sqrt{1+2\gamma t/\alpha}\right)}{K_1(\alpha)\sqrt{1+2\gamma t/\alpha}},\\ \int_0^\infty f(\nu)\,[\nu+\nu_h(\nu)]^m\, d\nu &= \gamma^m \frac{K_{1+m}(\alpha)}{K_1(\alpha)}.\end{aligned} \tag{23}$$

For $\nu_h = 0$ the solution is f_{H_∞}. In general, the left-hand side is the Laplace transform of the distribution of $s = \nu + \nu_h(\nu)$ induced by f, and the right-hand side is the Laplace transform of $f_{H_\infty}(s)$. By the uniqueness of the Laplace transform, Eq. (23) holds for all $t \geqslant 0$ only if the two distributions coincide. Every halo group has $s \geqslant \nu_h$, so if $\nu_h(\nu) \geqslant \nu_{h,\min} > 0$ for all ν, the induced distribution vanishes below $\nu_{h,\min}$, whereas f_{H_∞} is positive on $(0,\infty)$. Hence Eq. (23) has no solution $f \geqslant 0$ when the halo velocity dispersion is bounded away from zero. The best one can do is the push-forward solution

$$f(\nu) = f_{H_\infty}[\nu+\nu_h(\nu)]\left[1+\frac{d\nu_h}{d\nu}\right], \qquad D = \int_0^{\nu_h(0)} f_{H_\infty}(s)\, ds, \tag{24}$$

valid when $s(\nu)$ is increasing, which leaves the fraction D of the mixture unassigned. With the fits of Eq. (22), $s(\nu)$ is increasing ($ds/d\nu \geqslant 0.74$), $\nu_h(0) = 1.11$ and, for $\alpha = 1.33$ and $\gamma = 0.97$, $D = 0.27$. Figure 6 compares Eq. (24) with the simulation: for $0 < \ln\nu < 2$ it exceeds the simulation by a factor 1.27–1.47, while for $-4 < \ln\nu < -2$ it falls short by up to a factor of 12: the small halos are where the unassigned mass would have to go. This agrees with the direct test of Section 4.3. The maximum entropy X distribution and a halo population with $\sigma_h^2 \geqslant \sigma_{h,\min}^2 > 0$ are compatible only approximately, and a complete theory of the mass function needs a model of the halo velocity dispersion and of how the small halos share it.

5.2 Constant halo velocity dispersion and transformed functions

If $\nu_h \approx \lambda_h = \sigma_{hc}^2/\bar\sigma_h^2$ is taken constant, Eq. (23) becomes $\int_0^\infty f e^{-\nu t} d\nu \approx e^{\lambda_h t} K_1(\alpha\sqrt{1+2\gamma t/\alpha})/[K_1(\alpha)\sqrt{1+2\gamma t/\alpha}]$, so that the mass function is determined by the three parameters α, γ and λ_h. The same argument as above shows that this relation, too, can hold only approximately. For comparison, the moments and Laplace transforms of the PS, ST and double-λ functions are

$$\begin{aligned}\int_0^\infty f_{PS}\nu^k d\nu &= \frac{2^k\Gamma(1/2+k)}{\sqrt{\pi}},\\ \int_0^\infty f_{D\lambda}\nu^k d\nu &= \frac{(4\eta_0)^k\Gamma(q/2+k)}{\Gamma(q/2)},\\ \int_0^\infty f_{ST}\nu^k d\nu &= \left(\frac{2}{q}\right)^k \\ &\quad\times \frac{\Gamma(1/2+k)+2^{-p}\Gamma(1/2+k-p)}{\Gamma(1/2)+2^{-p}\Gamma(1/2-p)},\end{aligned} \tag{25}$$

$$\begin{aligned}&\int_0^\infty f_{PS}e^{-\nu t}d\nu = \frac{1}{\sqrt{1+2t}}, \qquad \int_0^\infty f_{D\lambda}e^{-\nu t}d\nu = (1+4\eta_0 t)^{-q/2},\\ &\int_0^\infty f_{ST}e^{-\nu t}d\nu = \sqrt{\frac{q}{q+2t}}\,\frac{\sqrt{\pi}+\Gamma(1/2-p)(1/2+t/q)^p}{\sqrt{\pi}+2^{-p}\Gamma(1/2-p)}.\end{aligned} \tag{26}$$

Figure 7 shows the transformed functions $F_X(t) = \int_0^\infty f_X(\nu)e^{-[\nu+\nu_h(\nu)]t}d\nu$ of the three forms, with ν_h from Eq. (22), against the right-hand side of Eq. (23) for $\alpha = 4/3$ and $\gamma = 0.8$, 0.85 and 0.9. The ST and double-λ forms nearly coincide and are closer to the target than PS. Note that γ depends on which σ_0 sets $v_0^2 = \sigma_0^2/3$: $\gamma = 0.78$ for $\sigma_0 = u_0$ and 0.97 for σ_0 of the halo particles.

6 CONCLUSIONS

The maximum entropy velocity distribution X of collisionless dark matter is a scale mixture of Gaussians, and its mixing distribution,

the distribution of the total velocity dispersion carried by the particles, is the generalized inverse Gaussian H_∞. This links the halo mass function, through the distribution H of the virial dispersion, to the maximum entropy distributions. Our conclusions are as follows.

(i) Without halo motion the mass function in $\nu = \sigma_v^2/\bar{\sigma}_h^2$ is f_{H_∞}. It is a maximum entropy distribution under normalisation, mean-dispersion and halo-number constraints for $n = 0$, but not for gravity.

(ii) For small halos, H reduces to the χ^2 form of PS with $\gamma = \alpha v_0^2/\bar{\sigma}_h^2 \approx 1$ (simulation: 0.97).

(iii) In the simulation, the Gaussian mixture built from the measured dispersions of the halo particles reproduces their velocity distribution to 0.03 dex. The mixing distribution, however, has a hard lower edge at $0.51\sigma_0^2$ set by the halo velocity dispersion, where the maximum entropy mixture has 0.28–0.34 of the particles.

(iv) Consequently, the complete equation for the mass function has no exact solution when the halo velocity dispersion is bounded away from zero; the push-forward solution leaves 0.27 of the mixture unassigned and underestimates the small halos by up to a factor of 12.

(v) The simulated mass function is described by the two-parameter P form (rms $= 0.18$ in $\ln f$) as well as by fitted double-λ and ST forms, but it is not redshift independent in ν: its mean shifts from 0.56 at $z = 2$ to 1.14 at $z = 0$.

A complete maximum entropy theory of the halo mass function therefore needs, besides the X distribution, a model of the halo velocity dispersion and of its share among the small halos; the random-walk description of the mass cascade [25, 26] provides one route.

DATA AVAILABILITY

Halo-based and correlation-based statistics of the simulation are available on Zenodo [37, 38], with the presentation slides [39] and all data files [40]. The reproducible package of this paper contains the manuscript source, the Python pipeline from the halo catalogue and the halo-group data to every figure and number, all results as CSV files, the author's original MATLAB scripts, an equation-by-equation audit and a verification log.

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APPENDIX A: STATISTICAL PROPERTIES OF H_∞ AND P

Here $J_s(x) = \int_s^x e^{-t}\sqrt{t^2-s^2}\,dt$ with $J_s(s) = 0$ and $J_s(\infty) = sK_1(s)$, $\Gamma(a,x)$ is the upper incomplete gamma function and $\gamma_E \approx 0.5772$ is Euler's constant. For X and Z (the speed distribution), $\int_0^\infty X(x)e^{xt}dx = \{v_0te^{-\alpha}(1+\alpha)+J_{\alpha\sqrt{1-(v_0t)^2}}(\alpha)\}/\{2\alpha K_1(\alpha)[1-(v_0t)^2]\} + K_1(\alpha\sqrt{1-(v_0t)^2})/\{2K_1(\alpha)\sqrt{1-(v_0t)^2}\}$ and the moment-generating function of Z is $2[P_X(t)+tP'_X(t)]$, with $P_X(t)$ the previous integral [27].

Table A1. Statistical properties of H_∞ (Eq. 14) and P. The moment-generating and characteristic functions of P are formal series: the moment-generating function of P does not exist for $t > 0$ (the series have zero radius of convergence). Every entry has been checked numerically.

	$H_\infty(x)$	$P(x)$
Support	$[0,\infty)$	$[0,\infty)$
PDF	$\frac{1}{2\alpha v_0^2 K_1(\alpha)} \exp\left[-\frac{\alpha}{2}\left(\frac{x}{\alpha v_0^2}+\frac{\alpha v_0^2}{x}\right)\right]$	$\frac{e^{-\sqrt{\alpha^2+x/v_0^2}}}{2\alpha v_0 K_1(\alpha)\sqrt{x}}$
CDF	$\frac{\alpha}{4K_1(\alpha)} \sum_{k=0}^{\infty} \frac{(-1)^k \alpha^{2k}}{k! 4^k} \Gamma\left(-1-k, \frac{\alpha^2 v_0^2}{2x}\right)$	$\frac{J_\alpha\left(\sqrt{\alpha^2+x/v_0^2}\right)+\sqrt{x/v_0^2}\, e^{-\sqrt{\alpha^2+x/v_0^2}}}{\alpha K_1(\alpha)}$
Mode	αv_0^2	0
Mean	$\frac{K_2(\alpha)}{K_1(\alpha)} \alpha v_0^2$	$\frac{K_2(\alpha)}{K_1(\alpha)} \alpha v_0^2$
Variance	$\left[\frac{K_3}{K_1} - \frac{K_2^2}{K_1^2}\right] (\alpha v_0^2)^2$	$\left[3\frac{K_3}{K_1} - \frac{K_2^2}{K_1^2}\right] (\alpha v_0^2)^2$
Moments	$\frac{K_{1+k}(\alpha)}{K_1(\alpha)} (\alpha v_0^2)^k$	$\frac{K_{1+k}(\alpha)}{K_1(\alpha)} \frac{\Gamma(k+1/2)}{\sqrt{\pi}} (2\alpha v_0^2)^k$
Generalized kurtosis (by the second moment)	$\frac{K_{1+k}}{K_1} \left(\frac{K_1}{K_3}\right)^{k/2}$	$\frac{K_{1+k}}{K_1} \left(\frac{K_1}{K_3}\right)^{k/2} \frac{\Gamma(k+1/2)}{\sqrt{\pi}} \left(\frac{2}{\sqrt{3}}\right)^k$
Entropy	$\alpha \frac{K_2}{K_1} - 1 + \ln[2\alpha v_0^2 K_1(\alpha)]$	$\ln[2\alpha v_0^2 K_1(\alpha)] + \left(\alpha + \frac{1}{2\alpha}\right) \frac{K_0}{K_1} - \frac{1}{2}(\gamma_E + \ln 2 - \ln\alpha) + 1$
Moment-generating function	$\frac{K_1\left(\alpha\sqrt{1-2v_0^2 t}\right)}{K_1(\alpha)\sqrt{1-2v_0^2 t}}$	$\sum_{k=0}^{\infty} \binom{2k}{k} \frac{K_{1+k}(\alpha)}{K_1(\alpha)} \left(\frac{\alpha v_0^2 t}{2}\right)^k$ (formal)
Characteristic function	$\frac{K_1\left(\alpha\sqrt{1-2iv_0^2 t}\right)}{K_1(\alpha)\sqrt{1-2iv_0^2 t}}$	$\sum_{k=0}^{\infty} \binom{2k}{k} \frac{K_{1+k}(\alpha)}{K_1(\alpha)} \left(\frac{i\alpha v_0^2 t}{2}\right)^k$ (formal)
Maximum entropy constraint	$\left\langle \frac{x}{2v_0^2} + \frac{\alpha^2 v_0^2}{2x} \right\rangle = \alpha \frac{K_2}{K_1} - 1$	